AeroAuto: Multi-Element Wing Optimization Framework

Theoretical Foundations, Computational Methodology, and Operational Guide

Applied Aerodynamic Design for Formula Student (FSAE / FSUK / FSG)

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Abstract

This engineering document provides an exhaustive theoretical manual and practical operational guide for using **AeroAuto** to design, analyze, and optimize multi-element wings for Formula Student (FSAE/FSUK/FSG) racing vehicles. AeroAuto couples the JavaFoil high-order potential panel code and integral boundary layer formulations with automated parametric wing geometry generation, multi-threaded campaign sweeps, regulatory bounding-box compliance (FSUK 2026 and FSG v1.1), and numerical multi-objective local optimization. This manual covers low-Reynolds-number multi-element aerodynamics, A.M.O. Smith's interaction mechanisms, ground effect modeling, complete operational workflows across all seven software tabs, a step-by-step wing design playbook, and the downstream pipeline for exporting 2D optimized profile stacks to SolidWorks/CATIA and 3D CFD solvers.

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1 Introduction to Formula Student Aerodynamics

1.1 The Formula Student Aerodynamic Paradigm

Unlike commercial aviation or Formula 1, Formula Student racecars operate within an unusual aerodynamic regime characterized by three key constraints:

1. **Low Freestream Velocity**: Track speeds typically range between 10 m/s in hairpins and 28 m/s on the longest straightaways. The typical competition average velocity is approximately $15 - 18$ m/s ($54 - 65$ km/h).
2. **Low-to-Moderate Reynolds Numbers**: Given wing chord lengths between 0.25 m and 0.45 m, the chord Reynolds number resides between $Re = 300{,}000$ and $Re = 850{,}000$. In this regime, boundary layers are exceptionally prone to laminar separation bubbles, transitional instability, and catastrophic boundary layer detachment under adverse pressure gradients.
3. **Extreme Downforce Sensitivity**: Formula Student autocross and endurance tracks are dominated by tight, transient cornering maneuvers. Lap-time sensitivity analysis proves that lateral cornering acceleration ($a_y \approx \mu \cdot g$) contributes far more to lap performance than top-speed aerodynamic drag minimization. The aerodynamic figure of merit is therefore total downforce metric $C_L \cdot A$, while drag ($C_D \cdot A$) must be balanced against powertrain battery/fuel limits.

1.2 The Role of 2D Optimization with AeroAuto

Full 3D Computational Fluid Dynamics (CFD) with rotating wheels, chassis geometry, and wake interactions is computationally expensive. Attempting to optimize multi-element flap gaps, overlaps, and deflection angles directly in 3D CFD results in weeks of computing time with limited design space coverage.

AeroAuto solves this bottleneck by performing rapid 2D viscous panel analysis (via JavaFoil), exploring hundreds of candidate airfoil combinations, angles, and slot geometries in minutes. Once the optimal 2D sectional stack is isolated, the team exports clean DXF geometry directly into CAD for 3D extrusion, endplate integration, and final 3D CFD verification.

2 Aerodynamic Theory of Multi-Element Wings

2.1 A.M.O. Smith's Multi-Element Boundary Layer Mechanisms

A properly tuned multi-element wing stack can achieve section lift coefficients exceeding $C_L > 3.0$, far surpassing the maximum lift achievable by any isolated single airfoil ($C_{L,\max} \approx 1.8$). In 1975, A.M.O. Smith formulated the five foundational fluid dynamic mechanisms that govern this phenomenon:

1. **Slat / Flap Boundary Layer Re-initialization**: A single long chord forces the boundary layer to travel across a prolonged adverse pressure gradient, leading to exhaustion and separation. In a multi-element stack, each subsequent element starts with a brand new, thin laminar boundary layer capable of handling high adverse pressure gradients without detaching.
2. **Circulation Effect**: The bound circulation Γ generated by downstream flap elements induces an upwash velocity vector on the trailing edge of upstream elements. This increases the effective angle of attack of the forward element, intensifying its suction peak without inducing early stall.
3. **Dumping Effect**: The high-speed flow passing along the suction surface of a trailing flap element creates an off-body low-pressure region immediately behind the trailing edge of the forward element. This lowers the trailing-edge static pressure, relieving the severity of the required pressure recovery across the main element and suppressing separation.

4. **Off-the-Surface Pressure Recovery**: Viscous deceleration of the forward element's wake occurs in the free stream above the flap rather than directly against a solid wall, permitting greater total pressure recovery without triggering wall separation.

5. **Fresh Boundary Layer Growth**: High-momentum air channeled through the slot nozzle injects kinetic energy into the near-wall region of the downstream flap, energizing its boundary layer.

2.2 Critical Slot Geometry Parameters

The interaction between elements is extraordinarily sensitive to relative spatial positioning:

- **Slot Gap (g/c)**: The minimum orthogonal distance between the trailing edge of element i and the suction surface of element $i+1$. In AeroAuto, optimal gap ratios typically lie between 1.8% and 3.2% of main chord. If the gap is too narrow, viscous boundary layers merge, choking the slot. If too wide, circulation coupling and dumping effects diminish rapidly.

- **Slot Overlap (o/c)**: The horizontal overlap between the trailing edge of element i and the leading edge of element $i+1$. Optimal overlap typically ranges between 1.0% and 2.5%.

- **Deflection Angle (δ)**: The pitch angle of element $i+1$ relative to the main chord line. Typical high-performance stacks feature progressive deflection increases (e.g., Main 0°, Flap 1 $-12°$, Flap 2 $-28°$, Flap 3 $-44°$).

2.3 Ground Effect Physics (Front Wing Design)

The Front Wing operates in close proximity to the ground ($h/c \leq 0.35$). AeroAuto models this via the **Method of Images**, mirroring the multi-element vortex and source distributions across the ground plane ($y = 0$):

$$\Phi_{\text{total}}(x, y) = \Phi_{\text{wing}}(x, y) + \Phi_{\text{image}}(x, -y) \tag{1}$$

This satisfies the zero-normal-flux boundary condition $v_y(x, 0) = 0$. The ground creates a converging-diverging venturi channel between the road and the wing underside, dramatically increasing flow speed, dropping static pressure, and producing massive ground-effect downforce. However, if the main element angle of attack is too steep or ride height too low, the channel boundary layer detaches, causing sudden *ground stall*.

3 AeroAuto Software Architecture Operations

AeroAuto is organized into seven specialized modules accessible via the top tab bar:

3.1 Tab 1: Configure

- **Main Element**: Select airfoil profile from the library (e.g., S1223, E423).

- **Global Parameters**: Set Chord (c in mm), Velocity (V in m/s), Reynolds Number (Re), and Angle of Attack sweep parameters ($\alpha_0, \alpha_1, \Delta\alpha$).

- **Flap Array (F1, F2, F3)**: Independent controls for each flap: Profile, Chord Scale ($S\%$), Deflection Angle ($A°$), X Position ($X\%$), Y Position ($Y\%$), and Active Toggle.

- **Regulation Ground Effect**: Toggle FSUK 2026 / FSG v1.1 bounding-box validation; enable Ground Effect with height percentage ($h/c\%$).

- **Sanity Check**: Instant computation of bounding box dimensions, ground clearance, slot gap percentages, and overlap percentages.

- **Interactive Preview**: Matplotlib preview with element color coding, ground plane, and hover coordinate tooltips.
- **Simulate**: Launches instant single JavaFoil simulation or opens in terminal.

3.2 Tab 2: Airfoils

- Built-in airfoil database containing high-lift low-Re profiles (S1223, E423, FX 63-137, CH10, etc.).
- Import external custom coordinates in Selig or Lednicer '.dat' format.
- Automatic coordinate normalization, point resampling, and thickness/camber curve visualization.

3.3 Tab 3: Campaigns

- Autonomous multi-threaded parameter sweeps leveraging CPU worker threads.
- Explores permutations across main profiles, flap profiles, scale ratios, deflections, and slot offsets.
- Evaluates both Unbounded Flow and Ground Effect flow regimes.
- Live Pareto frontier scatter plot ($C_L \cdot A$ vs $C_D \cdot A$) and progress ETA tracking.

3.4 Tab 4: Results

- Dataframe inspector presenting 56 physical and regulatory parameters per simulation run.
- Sortable columns: Downforce (N), Drag (N), L/D, $C_L \cdot A$, $C_D \cdot A$, Power Penalty (kW at speed), Stack Height, Stack Length, Ground Clearance, and Pitch Sensitivity (ΔC_L).
- Regulatory Filter: Isolate rule-compliant setups with a single toggle.
- **CAD Export**: Direct 1:1 millimeter scale export to DXF and SVG.

3.5 Tab 5: Graphs

- Publication-grade aerodynamic polar curves ($C_L, C_D, C_M, L/D$ vs α).
- Surface pressure distribution (C_p vs x/c) for every element in the stack.
- Boundary layer diagnostics: Skin friction coefficient (C_f), displacement thickness (δ^*), momentum thickness (θ), and separation point markers.

3.6 Tab 6: Optimizer

- Multi-objective numerical optimization engine utilizing JavaFoil simulation feedback.
- Custom objective weighting: Maximize Downforce, Maximize Efficiency, Minimize Drag, Maximize Pitch Stability, Minimize Stack Envelope.
- Automated geometric perturbation of flap offsets converging toward optimal slot geometry.
- One-click injection of optimized configurations back into the Configure tab.

3.7 Tab 7: Themes Settings

- Visual appearance styles (Dark, Light, High Contrast).
- Tooltip delay, wrap width, and navigation shortcuts.

4 The FSAE Wing Design Playbook: Step-by-Step

4.1 Step 1: Front Wing (FW) Optimization

1. Select role **Front Wing** (Span = 1400 mm, ground clearance limits active).
2. Enable **Ground Effect** with height set to $h/c \approx 25\% - 35\%$ (physical height $30 - 45$ mm).
3. Choose a high-camber profile with a generous leading-edge radius (e.g., `s1223`).
4. Set Main element angle to a conservative pitch ($0°$ to $-4°$) to preserve venturi channel integrity during pitch changes.
5. Configure Flap 1: $S = 30\%$, Angle $-12°$, Gap 2.2%, Overlap 1.5%.
6. Configure Flap 2: $S = 22\%$, Angle $-26°$, Gap 2.0%, Overlap 1.2%.
7. Run **Sanity Check**: Ensure overall stack length ≤ 500 mm and ground clearance ≥ 30 mm.
8. Check Pitch Sensitivity: Compare $C_{L,\alpha=-2°}$ vs $C_{L,\alpha=+2°}$. Under hard braking, the front wing dips; the slope $dC_L/d\alpha$ must remain stable and non-stalling.

4.2 Step 2: Rear Wing (RW) Optimization

1. Select role **Rear Wing** (Span = 950 mm, Unbounded free-stream flow).
2. Because the aspect ratio is relatively low ($AR = b^2/S \approx 2.5 - 3.2$), induced drag is substantial.
3. Deploy an aggressive 3-element or 4-element stack:
   - Main element: Chord $300 - 340$ mm, Angle $-6°$.
   - Flap 1: $S = 35\%$, Angle $-20°$, Gap 2.4%, Overlap 1.8%.
   - Flap 2: $S = 25\%$, Angle $-36°$, Gap 2.2%, Overlap 1.5%.
   - Flap 3: $S = 18\%$, Angle $-52°$, Gap 2.0%, Overlap 1.2%.
4. Target Metrics: Maximize $C_L \cdot A \geq 1.85\,\text{m}^2$ while ensuring $C_D \cdot A \leq 0.95\,\text{m}^2$.
5. Check Power Penalty: Ensure drag power at 20 m/s does not exceed the vehicle's electrical or engine budget (< 4.5 kW).

5 From AeroAuto 2D to 3D CAD and CFD Pipeline

1. **Export Geometry**: In Tab 4 (Results), click `Export DXF`.
2. **CAD Import**: In SolidWorks, CATIA, or NX, import the DXF onto the wing reference plane.
3. **Extrusion Endplates**:
   - Extrude profiles across half-span (700 mm for FW, 475 mm for RW).
   - Model full-depth endplates to prevent high-pressure air from wrapping into the suction side, preserving 2D sectional lift efficiency.
4. **3D CFD Verification**: Run steady-state RANS ($k - \omega$ SST turbulence model) with rotating wheels and ground velocity to capture 3D vortex structures and tire wake interactions.

6 Defending Your Aerodynamic Design in Competition

When presenting your aerodynamic development to Formula Student design judges:

- **Show Design Space Exploration**: Display AeroAuto campaign scatter plots proving that your team evaluated over 500 candidate stacks before choosing the final geometry.
- **Explain the Fluid Mechanics**: Use Tab 5 C_p curves to explain how flap slot flow suppressed the main element trailing-edge adverse pressure gradient.
- **Highlight Pitch Stability**: Emphasize that your wing was optimized not just for peak downforce at $\alpha = 0^\circ$, but maintained stable balance across $\pm 2^\circ$ pitch variations during braking and cornering.